

Kokpit kontroli wykorzystania budżetu

Budget Control Cockpit: an interactive controlling dashboard built with Power Pivot and DAX in Microsoft Excel

Microsoft Excel · Power Pivot · DAX · Data Modelling · Financial Controlling | File: Kokpit_Budzetu.xlsx | Final grade: 5.0 (A)

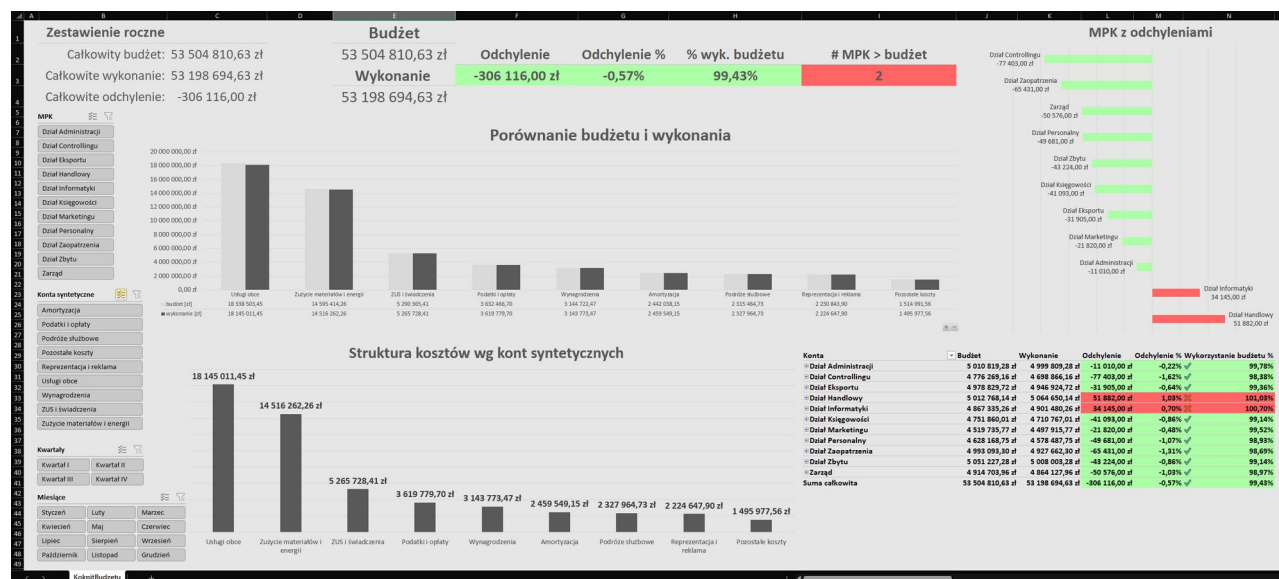


Figure 1. The cockpit in its default state: all 11 cost centres, full year

1. Project overview

A one-screen controlling cockpit that shows how 11 cost centres (MPK, Miejsca Powstawania Kosztów) used their annual budget. Managers and controllers can compare budget against actual spending for the whole organisation, a single department, one cost category or one month, and drill down from high-level cost groups to individual analytical accounts.

The project was the final assignment for a Power Pivot course. It met every mandatory requirement, added analytical features beyond the brief, and received the top grade (5.0).

Business questions the cockpit answers

- Is the organisation, a department or a cost category over or under budget, and by how much (in PLN and in %)?
- Which cost centres overspent and which saved money, and how many cost centres are over budget right now?
- Which synthetic cost groups, and which analytical accounts inside them, cause the deviations?
- How does the picture change by quarter or month?

2. Source data

The source was an Excel workbook of real budget and spending data covering 12 months of 2025:

Item	Scope
Transactions	21,346 records (10,673 budget + 10,673 actual)
Cost centres (MPK)	11 departments, e.g. Dział Informatyki, Dział Marketingu, Dział Handlowy, Zarząd

Item	Scope
Synthetic accounts	9 parent cost groups, e.g. Usługi obce, Zużycie materiałów i energii, Wynagrodzenia
Analytical accounts	72 detailed accounts, e.g. PFRON, Podatek VAT, Kursy i Szkolenia
Annual totals	Budget 53,504,810.63 PLN · Actual 53,198,694.63 PLN · Saving 306,116.00 PLN (−0.57%)

3. Data model (Power Pivot)

I imported four tables into the Power Pivot data model and arranged them as a star schema: one fact table and three dimension tables, joined by one-to-many relationships from each dimension to the fact table. Before loading, I checked data types and made sure the dimension keys contained no duplicates.

Table	Role	Key columns
dane	Fact	numer konta analitycznego, numer konta syntetycznego, numer MPK, m-c, rodzaj danych (budżet / wykonanie), kwota
MPK	Dimension	numer MPK → Nazwa MPK
konta syntetyczne	Dimension	numer konta syntetycznego → nazwa
konta analityczne	Dimension	numer konta analitycznego → nazwa

Calculated columns

I added two DAX calculated columns so the data could be filtered by time using readable Polish labels:

```
miesiąc =
PATHITEM(
"Styczeń|Luty|Marzec|Kwiecień|Maj|Czerwiec|Lipiec|Sierpień|Wrzesień|Październik|
Listopad|Grudzień";
[m-c]
)

kwartał =
IF(
'dane'[m-c] <= 3; "Kwartał I";
IF(
'dane'[m-c] <= 6; "Kwartał II";
IF(
'dane'[m-c] <= 9; "Kwartał III";
"Kwartał IV"
)
)
)
```

Hierarchies

- **KOSZTY** (required): synthetic account → analytical account. Supports drill-down and drill-up in the pivot table and pivot chart.
- **KALENDARZ** (added beyond the brief): quarter → month. Supports time-based analysis and the Quarters slicer.

4. DAX measures

All calculations are Power Pivot measures written in DAX. The dashboard contains no worksheet formulas. Measures are formatted in the model as PLN currency, percentages or whole numbers, so every pivot table displays them the same way.

Core measures (required)

```
budżet:=CALCULATE(SUM(dane[kwota]);dane[rodzaj_danych]="budżet")
wykonanie:=CALCULATE(SUM(dane[kwota]);dane[rodzaj_danych]="wykonanie")
odchylenie:=[wykonanie]-[budżet]
odchylenie%:=DIVIDE([odchylenie];[budżet];0)
wykorzystanie_budżetu%:=DIVIDE([wykonanie];[budżet];0)
```

Additional measures (beyond the brief)

```
liczba_MPK_z_przekroczeniem:=VAR wynik =
SUMX(MPK; IF([wykonanie] > [budżet]; 1; 0))
RETURN
IF(wynik = 0; BLANK(); wynik)

całkowity_budżet_roczny:=CALCULATE([budżet]; ALL('dane'))
całkowite_wykonanie_roczne:=CALCULATE([wykonanie]; ALL('dane'))
całkowite_odchylenie_roczne:=CALCULATE([odchylenie]; ALL('dane'))
```

Why they matter: the over-budget counter iterates over the MPK table and checks each cost centre separately, so it stays correct under any combination of slicers. The annual measures use ALL() to ignore slicer filters, so the "Zestawienie roczne" panel always shows the full-year baseline next to the filtered figures.

5. Dashboard design

The whole cockpit sits on one worksheet (KokpitBudżetu) and fits on one screen without scrolling. It contains 11 pivot tables and 3 pivot charts. All of them are connected to the same slicers, so a single click updates every visual.

Component	What it shows
Annual summary	Full-year budget, actual and deviation (unaffected by slicers) as a fixed reference point
KPI tiles	Budget, Actual, Deviation, Deviation %, % of budget used, # of cost centres over budget. Conditional formatting colours each tile green (saving) or red (overspend)
Budget vs. actual column chart	Clustered pivot chart built on the KOSZTY hierarchy with a data table below. Clicking + drills from synthetic to analytical accounts
Cost structure chart	Actual spending by synthetic account, sorted from largest to smallest, with data labels
Cost centres with deviations	Horizontal bar chart of each cost centre's deviation: green bars for savings, red bars for overspending
Main pivot table	Cost centre → KOSZTY hierarchy with Budget, Actual, Deviation, Deviation % and Budget use %. Green/red cell colouring plus ✓ / ✗ icon sets
Slicers	Cost centre (MPK), Synthetic account, Quarter, Month

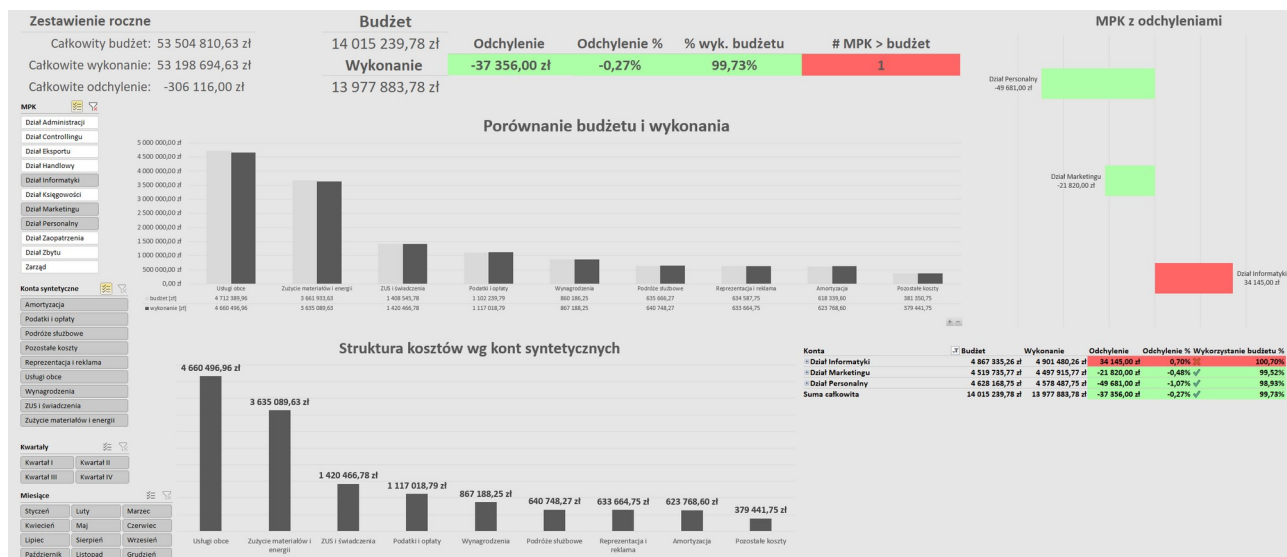


Figure 2. Filtered to three departments: the KPIs, charts and table all recalculate together

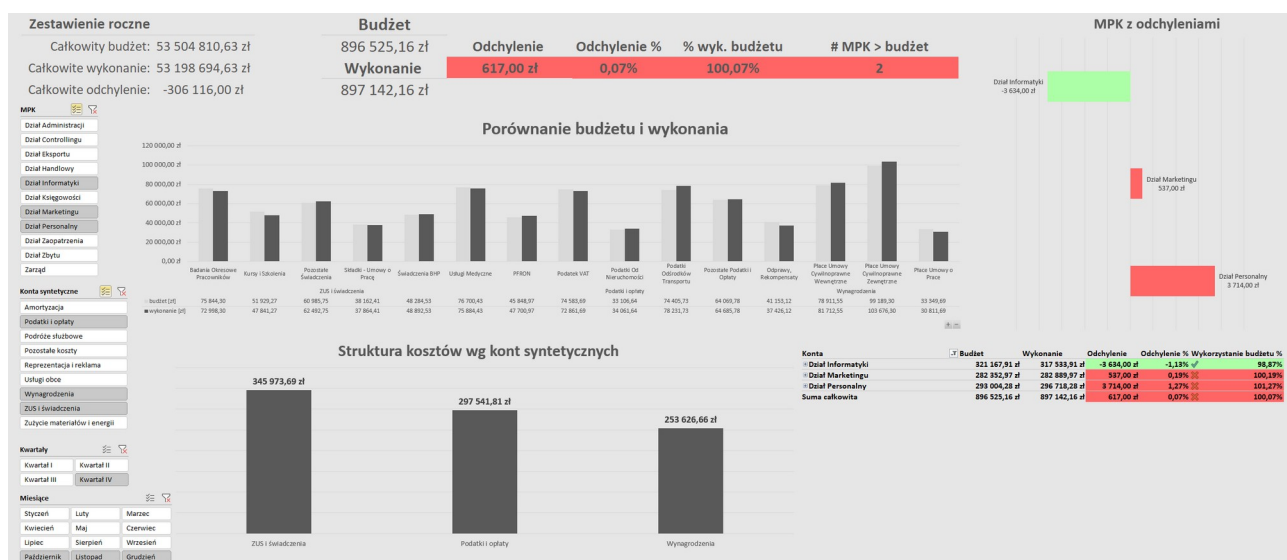


Figure 3. Q4 with analytical-account drill-down: the headline figure turns red (+0.07%) even though Dział Informatyki is under budget

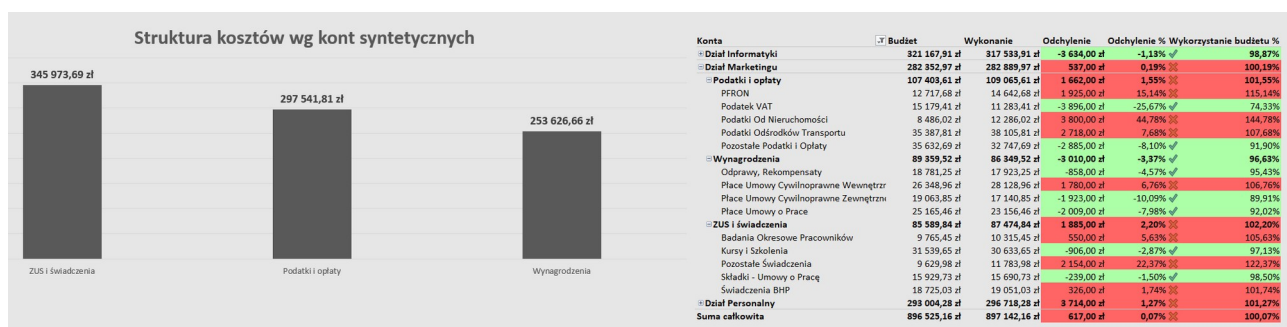


Figure 4. Drill-down in the main pivot table: cost centre → synthetic account → analytical account

6. Requirements vs. delivery

Requirement from the brief	Status	Notes
Data model in Power Pivot with relationships and correct data types	Met	Star schema with 3 one-to-many relationships
KOSZTY hierarchy (synthetic → analytical)	Met	Used in the main pivot table and the column chart

Requirement from the brief	Status	Notes
DAX measures: Budżet, Wykonanie, Odchylenie, Odchylenie %, Wykorzystanie %	Met	All written in DAX with number formats set in the model
Main pivot table with drill-down	Met	Nested under cost centres with conditional formatting and icons
Column chart with drill-down	Met	Pivot chart on the KOSZTY hierarchy with a data table
Pie or bar chart of cost structure	Met	Sorted column chart with data labels
Cost centres with deviations (overspend and savings)	Met	Bar chart coloured green/red
KPI indicators	Exceeded	6 dynamic KPI tiles plus a fixed annual summary
Slicers: cost centre, month, synthetic account	Exceeded	Quarter slicer and KALENDARZ hierarchy added
Red/green conditional formatting	Met	Applied to KPI tiles, the table and icon sets
Everything calculated in DAX, fully interactive, fits on one screen	Met	No worksheet formulas; every visual is connected to all slicers

7. Key insights from the data

- For the full year, the organisation spent **99.43%** of its budget, a saving of 306,116 PLN.
- **2 of 11** cost centres went over budget: Dział Handlowy (+51,882 PLN, 101.03%) and Dział Informatyki (+34,145 PLN, 100.70%).
- Dział Controllingu saved the most (-77,403 PLN, -1.62%).
- Usługi obce (≈18.1 M PLN) and Zużycie materiałów i energii (≈14.5 M PLN) together make up more than 60% of total spending.
- A small total deviation can hide large ones at account level. In Q4, Dział Marketingu's property tax account ran 44.78% over budget while its total deviation was only +0.19%. The drill-down makes this visible.

8. Skills and tools

Tools

- **Microsoft Excel:** pivot tables, pivot charts, slicers, conditional formatting, icon sets, dashboard layout
- **Power Pivot:** data model, relationships, hierarchies, calculated columns
- **DAX:** CALCULATE, DIVIDE, SUMX, ALL, PATHITEM, VAR/RETURN, BLANK

Skills

- Data modelling: star schema, fact and dimension tables, data-quality checks
- Building business measures and working with filter context
- Financial controlling: budget vs. actual variance analysis at cost-centre level
- Dashboard design: KPI layout, visual hierarchy, colour for meaning, one-screen ergonomics
- Interactive reporting: drill-down/drill-up and slicers connected to every visual
- Turning data into insight: finding the causes of deviations